SITE SAFETY

Early warning, Canada

OVERVIEW

Site
24-storey tower, Burnaby

Zones
10 work zones

Horizon
Next 5 work days

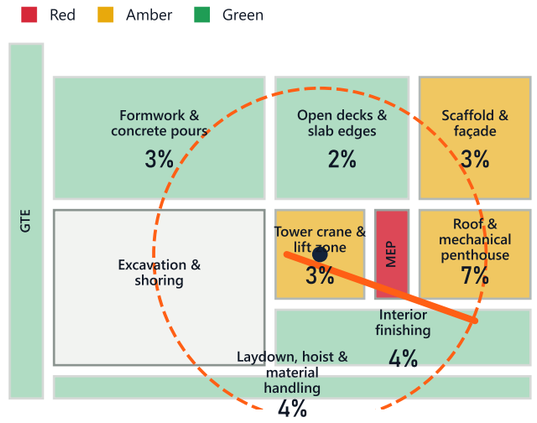
Scope
Burnaby tower, 10 zones

Analysis and design by
Osman Zakaria, PMP

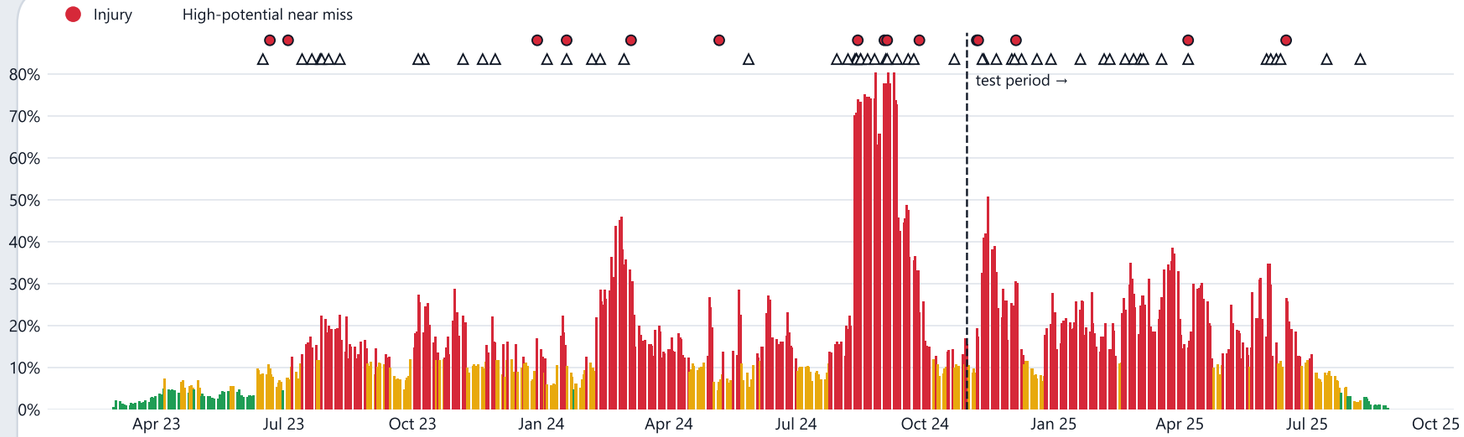
© 2026

84%	3.5×	5 days	0.75	2.8	2.7
Events flagged in advance (470)	Event rate on red zone-days	Median warning before an event	AUC (crew-only baseline 0.66)	Injuries / 100 person-yrs (BC: 3)	Red alerts per site-week

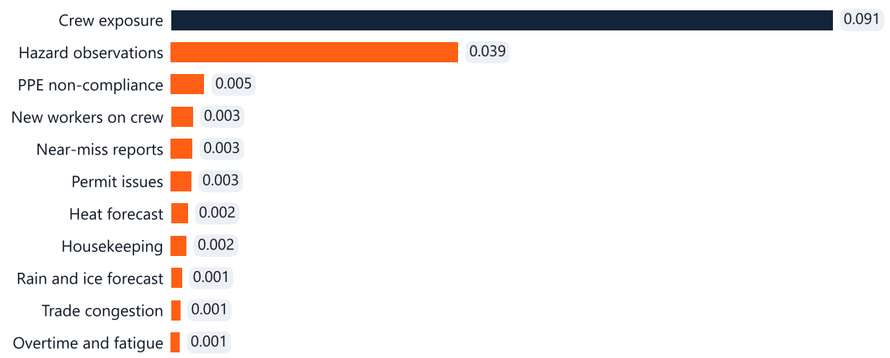
Site plan on the riskiest test day: Fri 15 Nov 2024



Highest zone risk on site each work day (5-day horizon), with injuries and near misses



What carries the warning: test AUC lost when each signal is shuffled



What the system does, and what it found

Answers where, not just how many

A schedule already says how many people will be on site. The model says which zone is building toward an injury in the next 5 work days and why, so the morning briefing can target it.

Warns early enough to act

84% of test-period events (and 81% of injuries) had an amber or red alert in the zone during the 5 work days before. Red zone-days carry 3.5 times the average event rate.

Without crying wolf

About 2.7 red and 10 amber zone-days per site each week, each with up to three drivers and controls tied to the BC OHS Regulation.

Grounded in BC numbers

The simulated portfolio runs at 2.8 time-loss injuries per 100 person-years against WorkSafeBC's 3 for general construction, with AWCBC's construction injury mix.

Sources: WorkSafeBC Statistics 2024; AWCBC NWISP 2020–2022; Alberta OHS PSI 2020; ECCC hourly climate (YVR). Independent work. Hypothetical sites with simulated daily logs, calibrated to WorkSafeBC, AWCBC, Statistics Canada and Alberta OHS data; real ECCC weather.

SITE SAFETY

Early warning, Canada

BC EVIDENCE

Site
24-storey tower, Burnaby

Zones
10 work zones

Horizon
Next 5 work days

Scope
WorkSafeBC, AWCBC, Alberta

Analysis and design by
Osman Zakaria, PMP

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316 — Construction deaths accepted, 2015–2024

13,864 — General-construction claims, 2024

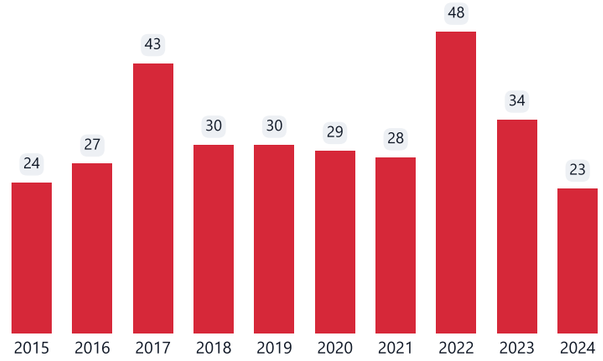
391k — Days lost to those claims

3 — Time-loss claims / 100 person-years

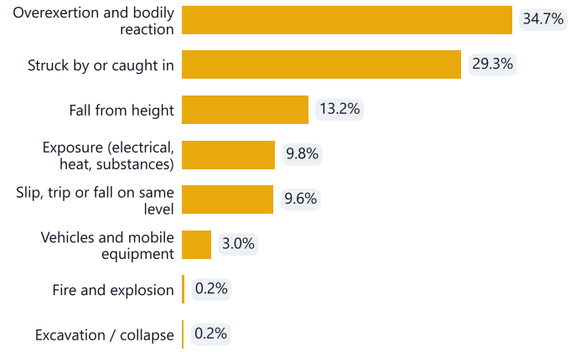
0.60 — Serious-injury rate, 2024

7,607 — BC construction lost-time claims, 2022

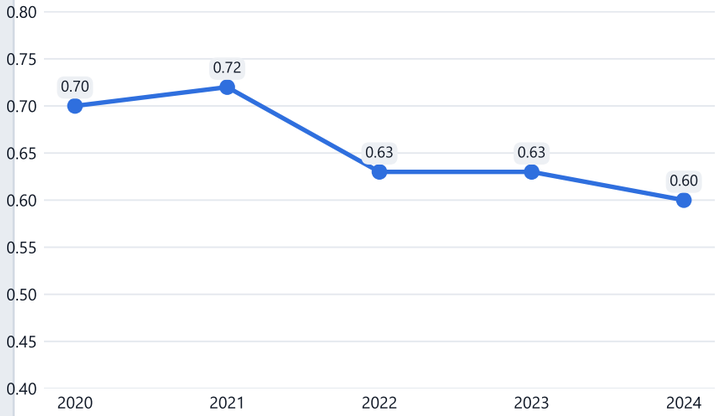
Work-related death claims, BC general construction



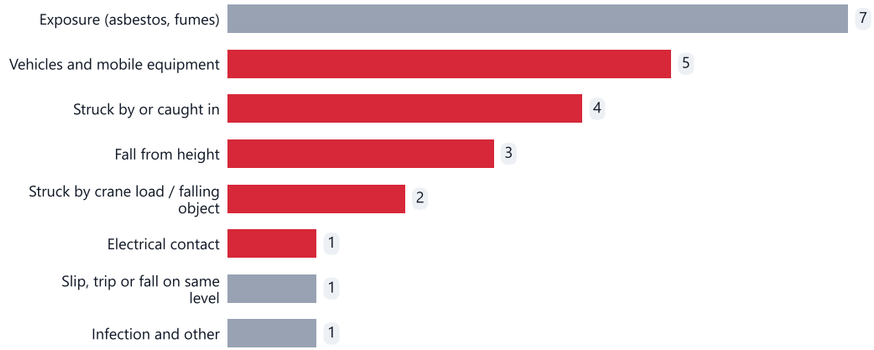
Lost-time claims in Canadian construction, by event, 2021–22



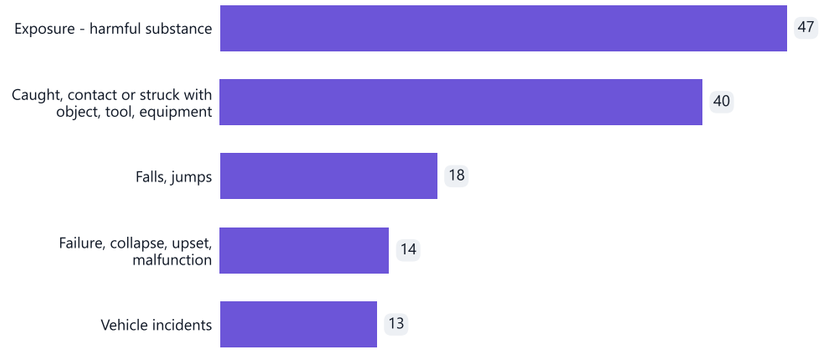
Serious-injury claims rate, BC construction (per 100 person-years)



BC construction deaths accepted in 2024, by hazard (WorkSafeBC descriptions)



Potentially serious incidents, Alberta construction 2020: 158 reported, 75% hurt no one



WorkSafeBC Statistics 2024 (Figures 1.4, 1.9, 2.7, 2.10; 2024 death descriptions); AWCBC NWISP 2020–2022 (Tables 15, 21); Alberta OHS PSI construction 2020. Independent work. Hypothetical sites with simulated daily logs, calibrated to WorkSafeBC, AWCBC, Statistics Canada and Alberta OHS data; real ECCC weather.

SITE SAFETY

Early warning, Canada

SITE & CALIBRATION

Site

24-storey tower, Burnaby

Zones

10 work zones

Horizon

Next 5 work days

Scope

30-month programme

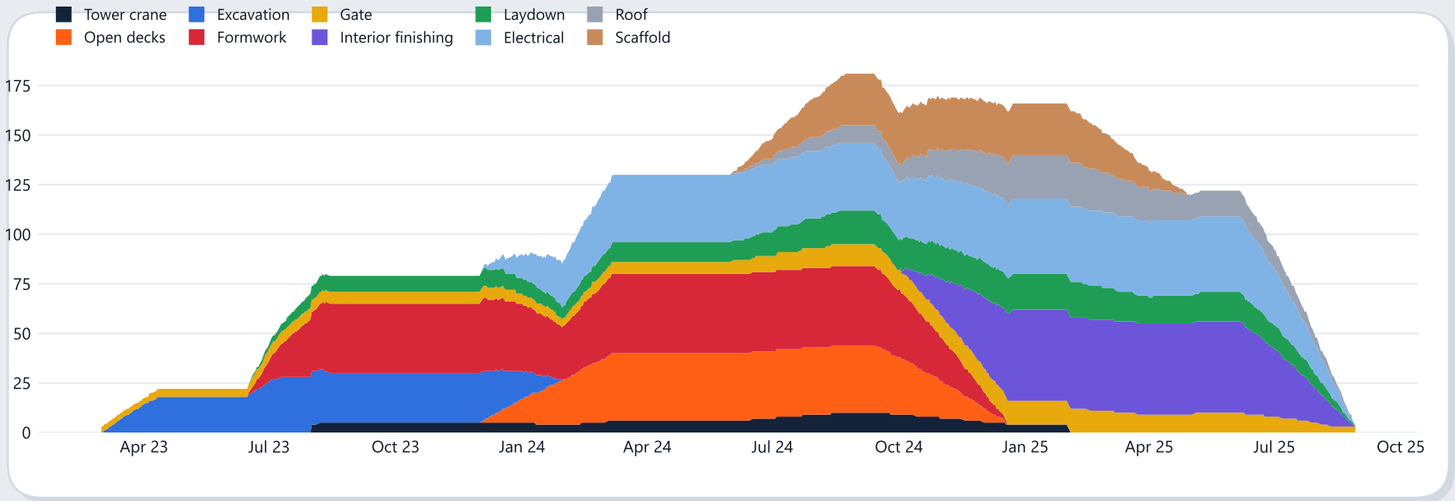
Analysis and design by

Osman Zakaria, PMP

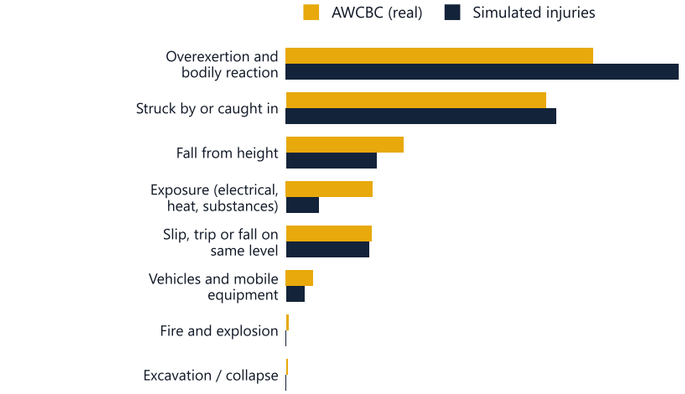
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181	13.7M	2.80	246	800	30 months
Peak workers on the Burnaby site	Person-hours, 30 sites	Time-loss rate (BC target 3)	Injuries across the portfolio	High-potential near misses	Programme, Mar 2023 – Aug 2025

Burnaby crews by zone across the programme (workers per day)



Simulated injury mix vs AWCBC construction mix



What is real

- **Base injury rate**

  BC general construction runs at 3 time-loss claims per 100 person-years (WorkSafeBC). Every site's hazard is rescaled so the portfolio matches it.

- **Injury mix and severity**

  Each injury takes an event family from AWCBC's construction lost-time claims by event, weighted by the zone's hazards; the serious share follows WorkSafeBC's serious-injury rate.

- **Near misses and weather**

  Three high-potential near misses per injury, after Alberta's 2020 construction PSIs (75% hurt no one); every day runs on real Vancouver shift weather from ECCC.

What is simulated, and why the model can't cheat

- **The site and its logs**

  Ten zones, a six-phase programme and daily logs: near-miss reports, hazard observations, PPE and housekeeping audits, permits, new workers, overtime and trade congestion.

- **A hidden safety climate**

  Each zone's climate drifts and sometimes deteriorates for weeks (a new subcontractor, a schedule crunch). It drives both the logs and the incidents, but the model never sees it.

- **Pooled across a portfolio**

  One site has too few injuries to learn from, so the model trains on a contractor's 30 sites in three provinces, as a real safety department would.

Crews follow a six-phase programme; injury rates are rescaled so the whole portfolio runs at WorkSafeBC's general-construction rate. Independent work. Hypothetical sites with simulated daily logs, calibrated to WorkSafeBC, AWCBC, Statistics Canada and Alberta OHS data; real ECCC weather.

SITE SAFETY

Early warning, Canada

DOES IT WORK?

Site
24-storey tower, Burnaby

Zones
10 work zones

Horizon
Next 5 work days

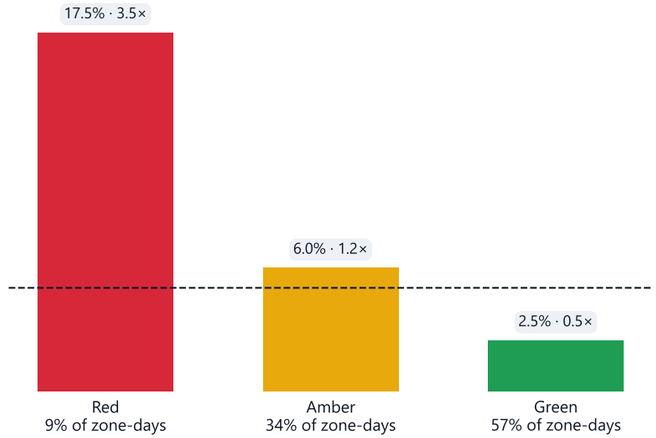
Scope
Test period, 30 sites

Analysis and design by
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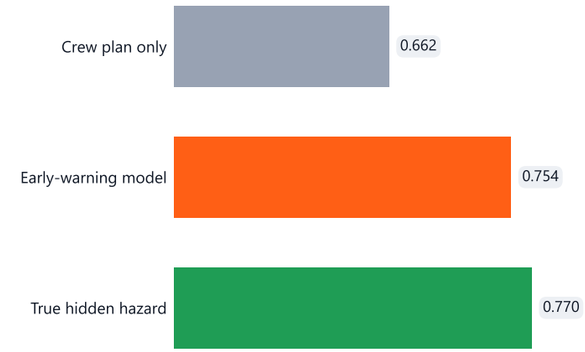
© 2026

0.75	0.66	0.77	0.17	3.5×	81%
AUC, test period	Crew-only baseline AUC	Ceiling: true hidden hazard	Average precision (base 5%)	Red-band lift	Injuries flagged in advance

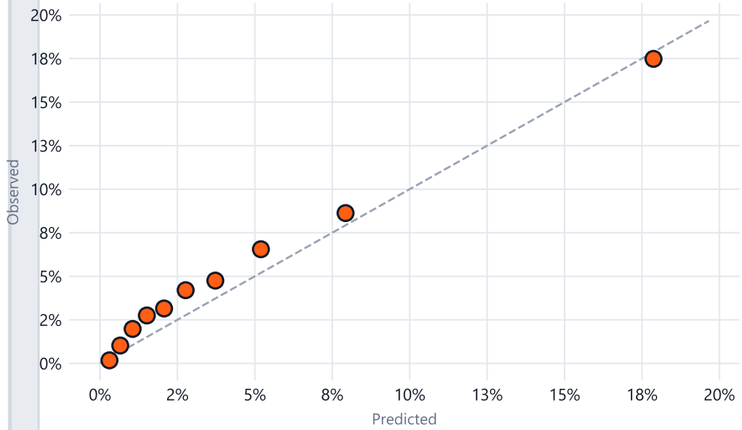
Event rate within 5 work days, by alert band



Ranking quality: model vs baseline vs ceiling (AUC)



Calibration by decile: predicted vs observed



Share of test-period events flagged amber or red in the 5 work days before



Reading the scorecard honestly

- **Better than the schedule alone**
  Knowing only the crew plan ranks zone-days at AUC 0.66; the leading indicators lift it to 0.75, closing about 86% of the gap to a model that could see the hidden hazard.
- **Rare events stay hard**
  With a 5% base rate most zone-days never see an event, so precision stays modest (0.17). That is why alerts drive cheap, targeted controls, not stop-work orders.
- **Simulated logs, real method**
  The indicators are simulated, so these numbers show what the method can do with data like this. A pilot on a real site's records is the next step.

Trained on all 30 sites (BC, ON, AB) before 2024-10-31; judged on the ten months after. Bands fixed on the training period. Independent work. Hypothetical sites with simulated daily logs, calibrated to WorkSafeBC, AWCBC, Statistics Canada and Alberta OHS data; real ECCC weather.

SITE SAFETY

Early warning, Canada

WARNING RECORD

Site
24-storey tower, Burnaby

Zones
10 work zones

Horizon
Next 5 work days

Scope
Zone by zone, week by week

Analysis and design by
Osman Zakaria, PMP

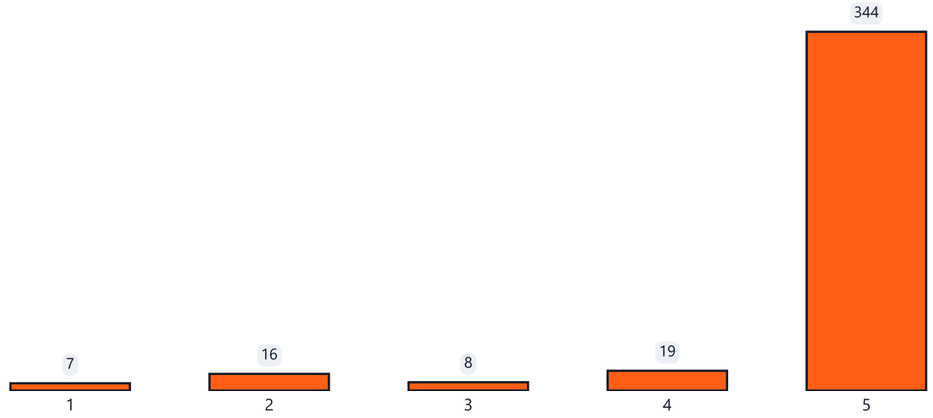
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5 days	84%	491	1209	72	470
Median warning lead	Test events with prior warning	Red zone-days, Burnaby	Amber zone-days, Burnaby	Burnaby events, whole build	Test events, all sites

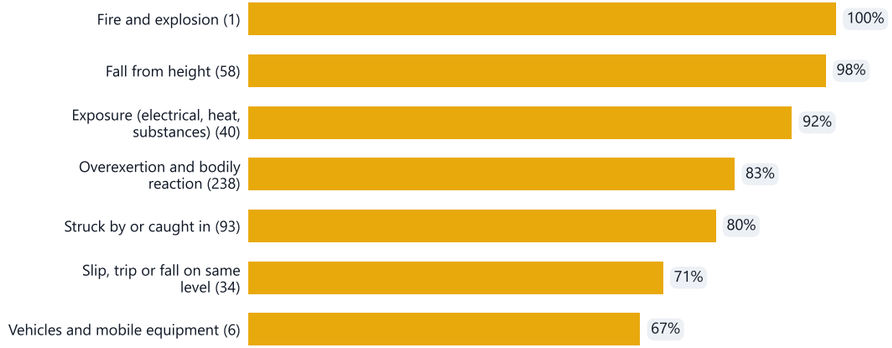
Weekly peak alert by zone, Burnaby, March 2023 – August 2025



Warning lead time for flagged test events (work days before the event)



Test events by hazard: share flagged in advance



Weekly peak alert per zone for the Burnaby build; dots are that week's injuries (filled) and high-potential near misses (open). Independent work. Hypothetical sites with simulated daily logs, calibrated to WorkSafeBC, AWCBC, Statistics Canada and Alberta OHS data; real ECCC weather.

SITE SAFETY

Early warning, Canada

WEATHER TRIGGERS

Site
24-storey tower, Burnaby

Zones
10 work zones

Horizon
Next 5 work days

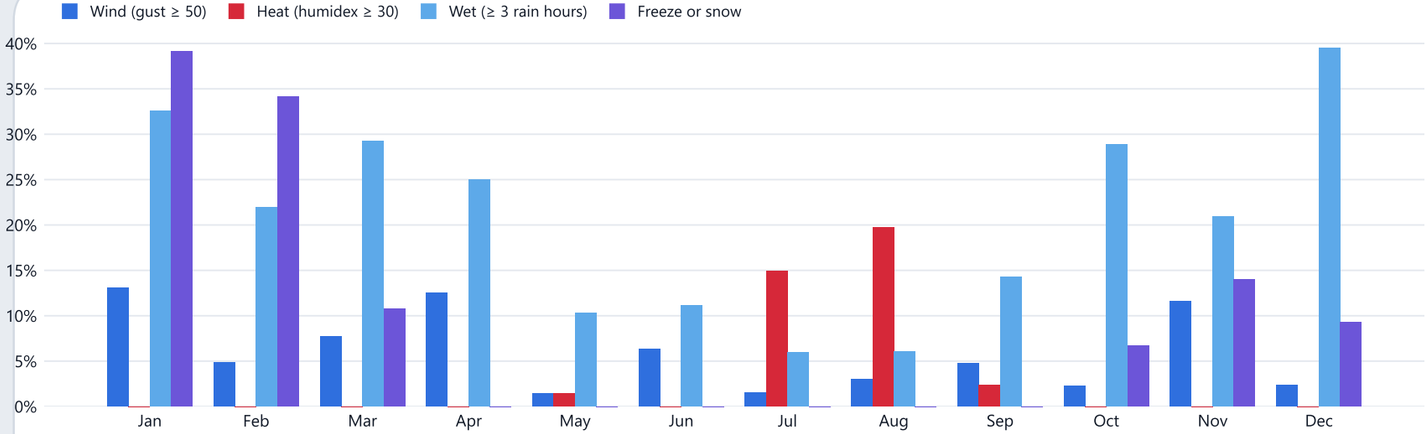
Scope
Real YVR shift weather

Analysis and design by
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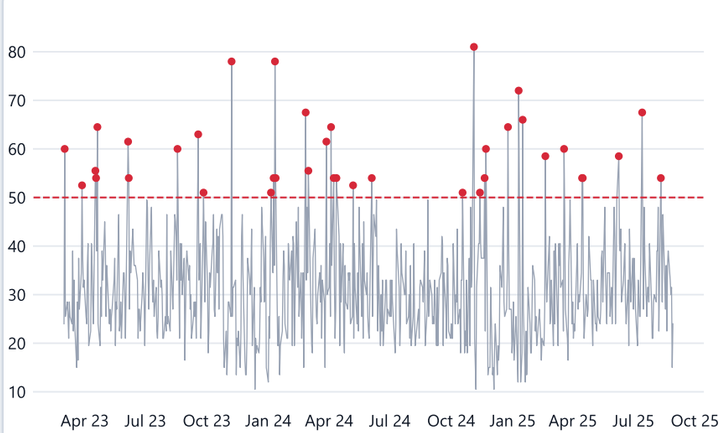
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38	25	126	52	81 km/h	653
Crane wind stand-down days	Heat-stress screening days	Wet-deck days (3+ rain hours)	Freeze or snow days	Highest estimated gust	Work days of real YVR weather

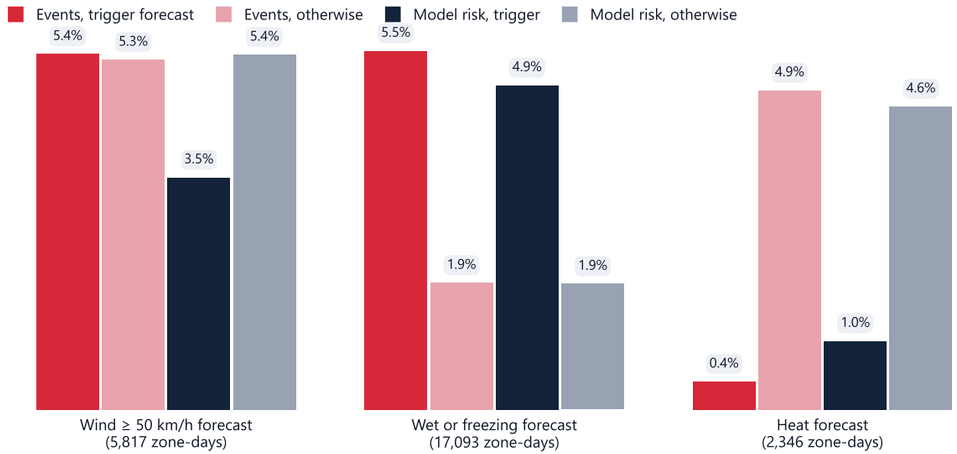
Weather triggers by month (share of work days), Vancouver 2023–2025



Estimated gust through the build (crane stand-down at 50 km/h)



Exposed zones, test period: actual event rate and model risk when the 5-day forecast has a trigger



What the weather check found, and the fix

Rain and frost: learned

When wet or freezing days are forecast, exposed zones see 2.8× the event rate and the model raises their risk 2.6×. It has seen enough wet winters to weigh this correctly.

Wind: under-learned, so a hard rule backs it up

Events run 1.0× higher in windy windows, but wind incidents are too rare for the model to weigh, so its risk barely moves. A rule now sets crane, scaffold and roof zones to at least amber whenever a gust of 50 km/h or more is forecast.

Heat: too little evidence yet

The test period's heat fell in summer 2025, after the roof, deck and formwork crews had wound down, so there are too few exposed zone-days to judge. Thresholds remain screens; manufacturers' limits and site procedures decide.

ECCC hourly observations at Vancouver Int'l A, 07:00–15:00, work days only. Gust estimated as 1.5 × the hourly mean wind. Independent work. Hypothetical sites with simulated daily logs, calibrated to WorkSafeBC, AWCBC, Statistics Canada and Alberta OHS data; real ECCC weather.

SITE SAFETY

Early warning, Canada

PROVINCES

Site
24-storey tower, Burnaby

Zones
10 work zones

Horizon
Next 5 work days

Scope
BC, Ontario, Alberta

1 2 3 4 5 6 7 8 9

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3 Provinces, 30 sites

3.3 BC claims / 100 workers

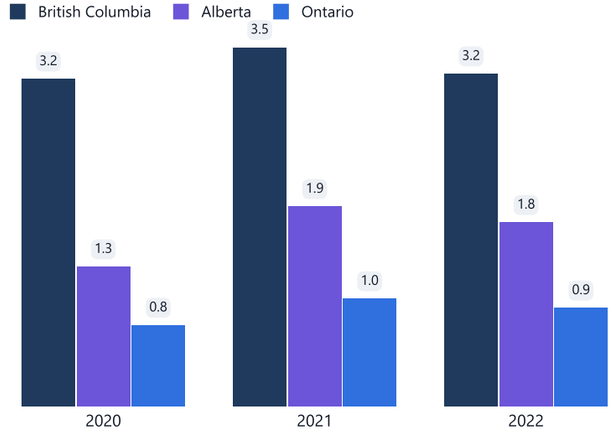
1.7 Alberta claims / 100 workers

0.9 Ontario claims / 100 workers

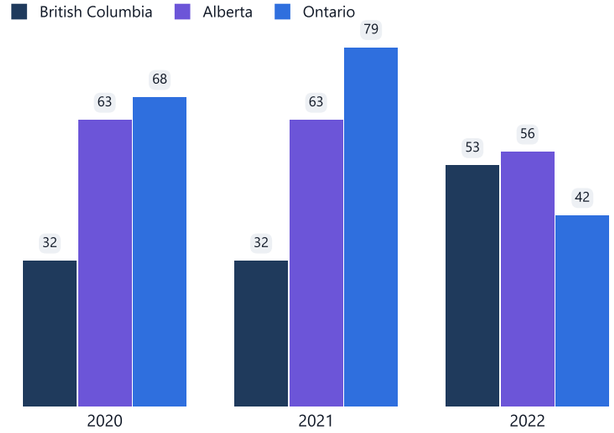
151 Construction deaths, 3 provinces, 2022

84% Test events flagged in advance

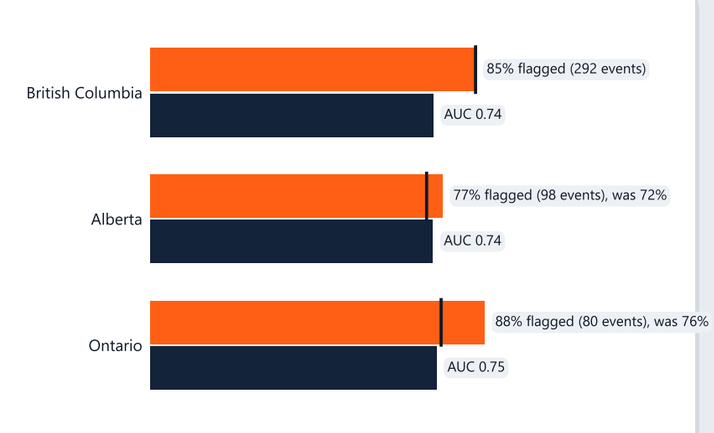
Construction lost-time claims per 100 workers



Construction fatalities accepted (AWCBC)



Flagged in advance with each province's own cut-off (| = shared cut-off)



Site scorecard (test period)

Site	Prov.	Events	Injuries	Flagged	Red days
Burnaby (featured)	BC	28	5	96%	209
Surrey	BC	18	7	89%	140
Richmond	BC	19	2	95%	204
Coquitlam	BC	15	3	73%	135
Vancouver	BC	32	8	91%	262
North Vancouver	BC	25	3	88%	132
New Westminster	BC	22	2	91%	233
Langley	BC	19	4	89%	106
Toronto	ON	12	7	83%	141
Mississauga	ON	8	3	88%	98
Vaughan	ON	8	1	75%	129
Markham	ON	11	3	91%	81
Calgary	AB	23	6	83%	166
Airdrie	AB	10	3	80%	130
Cochrane	AB	9	4	67%	57

Site	Prov.	Events	Injuries	Flagged	Red days
Kelowna	BC	25	2	84%	151
Victoria	BC	29	6	86%	158
Abbotsford	BC	22	4	68%	70
Port Moody	BC	12	1	42%	40
Delta	BC	16	6	88%	147
Maple Ridge	BC	10	2	90%	88
Ottawa	ON	14	6	100%	102
Hamilton	ON	7	0	100%	87
Brampton	ON	7	3	100%	39
Oakville	ON	5	3	60%	106
Kitchener	ON	8	2	75%	71
Edmonton	AB	20	7	75%	129
Red Deer	AB	10	4	80%	38
Lethbridge	AB	9	1	89%	44
St. Albert	AB	17	3	65%	126

Rates: AWCBC accepted construction lost-time claims ÷ Statistics Canada LFS construction employment (table 14-10-0023), 2020–2022. Boards differ in coverage and lost-time definitions. Independent work. Hypothetical sites with simulated daily logs, calibrated to WorkSafeBC, AWCBC, Statistics Canada and Alberta OHS data; real ECCC weather.

Site
24-storey tower, Burnaby

Zones
10 work zones

Horizon
Next 5 work days

Scope
OHS Regulation, data

Analysis and design by
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12 Driver-specific controls

10 Zone standing controls

5 Public data sources

BC·ON·AB Controls cited per province

Pre-shift When alerts are acted on

3 max Drivers shown per alert

Control library: what the briefing tells the crew to do, by driver

Near-miss reports
Stand-down toolbox talk on this zone's recent near misses; supervisor walks the zone before start
BC: OHSR Part 3: inspections and investigations · ON: OHSA s.9: joint health and safety committee

Hazard observations
Close out every open hazard observation before work starts; add a mid-shift inspection
BC: OHSR Part 3: workplace inspections · ON: OHSA s.9: workplace inspections

PPE non-compliance
PPE check at zone entry; replace damaged harnesses, eyewear and gloves
BC: OHSR Part 8: personal protective equipment · ON: O. Reg. 213/91: protective clothing and equipment

Housekeeping
Dedicated clean-up crew: clear debris, cords and offcuts from walkways and access routes
BC: OHSR Part 4: general conditions · ON: O. Reg. 213/91: housekeeping and access

New workers on crew
Pair every new worker with an experienced buddy; confirm site orientation is complete
BC: OHSR Part 3: young or new workers (3.22–3.25) · ON: O. Reg. 297/13: mandatory training

Overtime and fatigue
Cap overtime and rotate crews; fatigue check in the pre-shift meeting
BC: OHSR Part 4: impairment · ON: OHSA s.25(2)(h): employer general duty

Trade congestion
Re-sequence trades so one trade works each bay; settle conflicts at the coordination meeting
BC: Workers Compensation Act: prime contractor coordination · ON: OHSA s.23: constructor duties

Permit issues
Audit open permits (hot work, excavation, critical lifts) and stop work without a valid permit
BC: OHSR Parts 12, 14 and 20 · ON: O. Reg. 213/91: excavations, hoisting and hot work

Wind forecast
Stand down crane lifts above the manufacturer's wind limit; secure sheeting, panels and loose material
BC: OHSR Part 14: cranes and hoists; Part 13: scaffolds · ON: O. Reg. 213/91: cranes, hoisting and rigging

Heat forecast
Heat-stress plan: water, shade and work-rest cycles; start earlier and move heavy work to the morning
BC: OHSR Part 7: thermal exposure · ON: OHSA s.25(2)(h); ministry heat stress guidance

Rain and ice forecast
Sand or salt decks, ramps and ladders; check edge protection and anchors after the weather
BC: OHSR Part 4: general conditions; Part 11: fall protection · ON: O. Reg. 213/91: fall protection (s.26)

Crew exposure
Raise the supervision ratio for the peak crew; brief the day's highest-risk tasks
BC: OHSR Part 3: supervision · ON: OHSA s.27: supervisor duties

Data sources

Source	Provider	What it sets	Access
Statistics 2024	WorkSafeBC	BC construction time-loss and serious-injury rates, claims mix, deaths 2015–2024, 2024 death descriptions	worksafebc.com: Data & insights, statistical reports
National Work Injury, Disease and Fatality Statistics 2020–2022	AWCBC	Construction claims by event; claims and fatalities by province (BC, ON, AB)	awcbc.org: NWISP publications
Potentially Serious Incidents: construction, 2020	Government of Alberta, OHS	Near-miss ratio: 75% of construction PSIs hurt no one; PSI types	open.alberta.ca
Labour Force Survey, table 14-10-0023	Statistics Canada	Construction employment by province: turns AWCBC claims into rates for ON and AB	www150.statcan.gc.ca
Historical hourly climate: Vancouver, Toronto, Calgary	Environment and Climate Change Canada	Shift-hour wind, humidex, rain, frost and snow for every work day, 2023–2025	climate.weather.gc.ca bulk data

Controls cite BC's OHS Regulation, Ontario's OHSA and O. Reg. 213/91, and Alberta's OHS Act and Code (BC shown; all three on the live dashboard). They support, not replace, the site's program. Independent work. Hypothetical sites with simulated daily logs, calibrated to WorkSafeBC, AWCBC, Statistics Canada and Alberta OHS data; real ECCC weather.

SITE SAFETY

Early warning, Canada

REAL-WORLD CHECK

Sites
37,000 in New York City

Unit
One site, one week

Horizon
Next 4 weeks

Scope
Real public records

1 2 3 4 5 6 7 8 9

Analysis and design by
Osman Zakaria, PMP

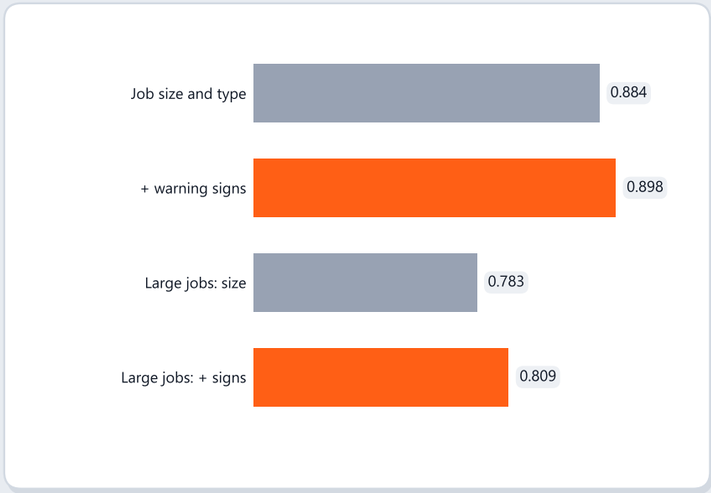
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86%	12×	0.90	0.81	37,000	4 weeks
Real incidents flagged (524)	Incident rate, red site-weeks	AUC (job size alone 0.88)	Large jobs (size alone 0.78)	NYC construction sites	Warning horizon

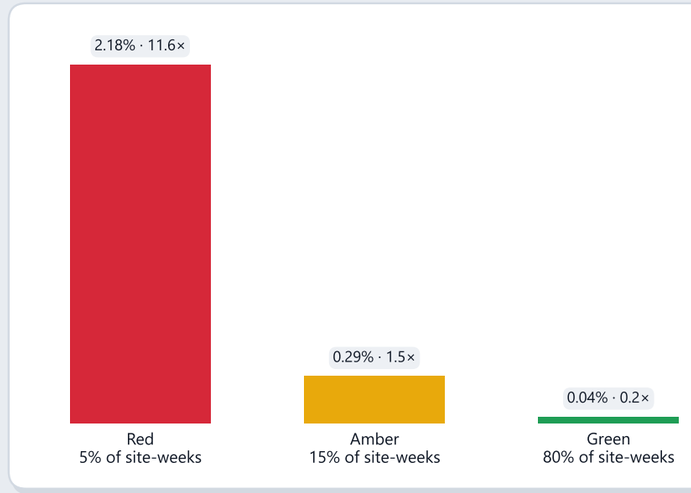
Large jobs: incident rate next 4 weeks, with vs without each sign



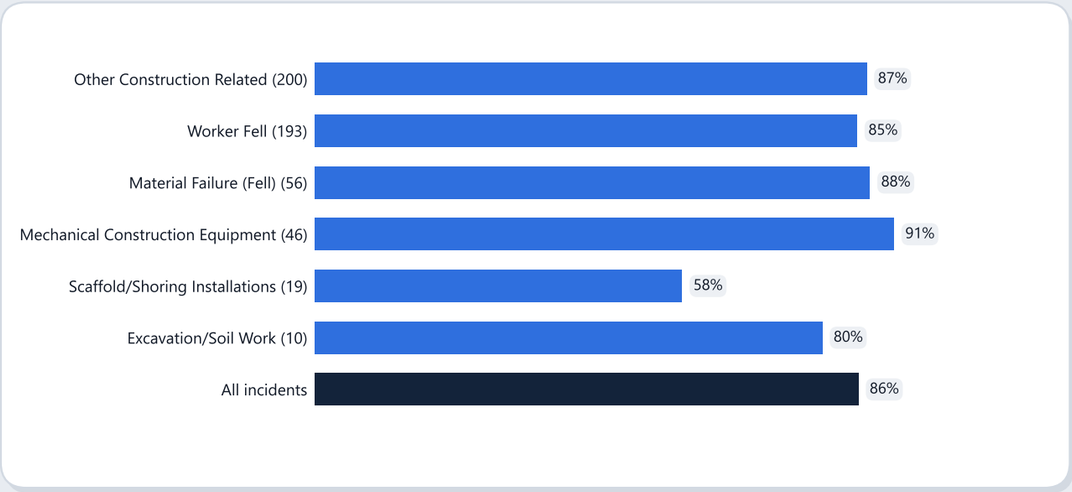
Ranking quality (AUC, test period)



Incident rate within 4 weeks, by alert band



Test-period incidents flagged amber or red in the 4 weeks before, by type



What the real-world check shows

- **It works on real sites**
  Alerts fixed on 2024 to mid-2025 caught 86% of 524 later accidents and incidents in the 4 weeks before; red site-weeks (5% of all) had 12× the average rate.
- **Exposure carries most of it**
  Job size and type alone rank sites at AUC 0.88. Big structural, excavation and demolition jobs are where incidents happen.
- **Warning signs still add**
  On large jobs they lift AUC from 0.78 to 0.81; a site with an incident in the past year was 10× as likely to have another within 4 weeks.
- **Not the zone model itself**
  Regulator records are sparse, site-level and partly reactive. A contractor pilot on daily logs is still the real test.

NYC Open Data, Department of Buildings: Construction-Related Incidents, DOB NOW approved permits, ECB violations, complaints. Trained on site-weeks 2024-01-01 to 2025-07-01; tested to 2026-09-28. Independent work. Real sites and real outcomes; nothing on this page is simulated.